Operating system and system programming (25CS2104E)

Process Terminating Messages

Review-I

Under the Esteemed Guidance of

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Abstract

Process Terminating Messages is a project focused on understanding how an operating system manages and communicates the termination of processes. A process may terminate normally after completing its task or abnormally due to errors, user actions, or system conditions. The project explains the different reasons for process termination and how the operating system handles the termination process. It also focuses on terminating messages, which provide information about why and how a process has ended. Understanding these messages helps in process management, error detection, debugging, and maintaining system stability. The project demonstrates the role of the operating system in safely terminating processes and communicating their status to users or other processes.

Introduction

In an operating system, a process goes through its life cycle, which includes creation, execution, waiting, and termination. Termination is the final stage, where the process exits the system and its resources are released for other processes. A process may terminate normally after completing its task or abnormally due to errors, exceptions, or external interventions.

When a process terminates, the operating system generates terminating messages to communicate the reason and status of termination. These messages are important for the parent process, the user, or other system components to understand what happened to the process. They help in error detection, debugging, resource management, and maintaining overall system stability.

This project explores the concept of process terminating messages, the different causes of process termination, and how the operating system handles and communicates termination information effectively.

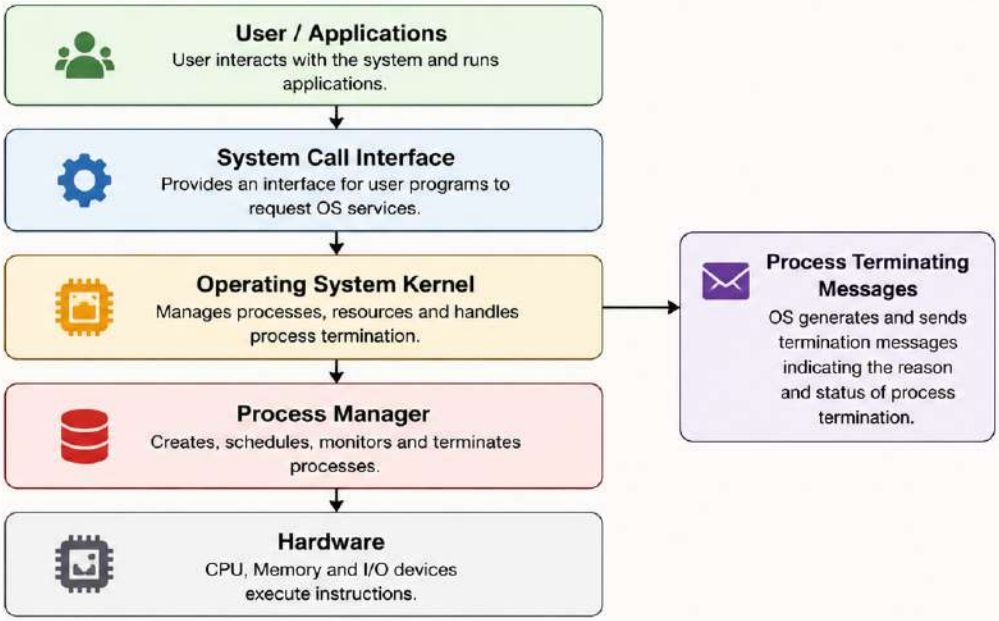
Hardware and Software requirements

List the Requirements in table format

Requirement Type	Requirements
Hardware Requirements	• Intel i3 or higher Processor • Minimum 4 GB RAM • Minimum 20 GB free hard disk space • Monitor, Keyboard and Mouse
Software Requirements	• Windows 10/11 or Linux (Ubuntu 20.04 or higher) • C / C++ Programming Language • GCC Compiler • Terminal / Command Prompt • Text Editor or IDE

System Architecture

The system architecture for Process Terminating Messages is shown below:



Thank you